

Excellent. Before ARINC 818-2, before Fibre Channel, before packets, before FPGA RTL, we must complete **Stage 1: Video Systems Fundamentals**.

Many FPGA engineers fail with ARINC 818 because they try to learn the protocol without understanding the thing being transported: **video**.

ARINC 818 is fundamentally a **video transport protocol**.

If you deeply understand video systems, 50% of ARINC 818 becomes intuitive.

STAGE 1 – VIDEO SYSTEMS FUNDAMENTALS

Objective

By the end of Stage 1, you should be able to:

- Explain how a video image is formed
- Calculate video bandwidth
- Calculate frame sizes
- Calculate pixel rates
- Understand display pipelines
- Understand image sensors
- Understand frame timing
- Understand synchronization
- Understand color formats
- Understand why ARINC 818 exists
- Understand what data actually flows through the link

Without referring to notes.

Module 1: What Is Video?

Most people answer:

Video is a moving image.

Technically wrong.

A video is:

A sequence of still images (frames) displayed at a sufficiently high rate to create the perception of motion.

Example:

30 frames/sec

Frame1

Frame2

Frame3

...

Frame30

Human eye perceives motion.

Important Concept:

Video = Frames + Timing

Not just frames.

Timing is equally important.

Module 2: What Is A Frame?

A frame is one complete image.

Example:

1920 × 1080

means

1920 pixels horizontally

1080 pixels vertically

Total pixels:

1920 × 1080

=

2,073,600 pixels

Approximately:

2.07 Megapixels

FPGA Perspective

A frame is simply:

Array of pixels

Example:

Pixel(0,0)

Pixel(0,1)

Pixel(0,2)

...

stored sequentially.

Must Understand

Frame is NOT transmitted as image.

It is transmitted as:

Pixel stream

Module 3: What Is A Pixel?

Pixel:

Picture Element

Smallest display element.

Monochrome Pixel

Example:

8-bit

Range:

0 → 255

0 = black

255 = white

Color Pixel

Usually:

Red

Green

Blue

components.

Example:

R = 255

G = 0

B = 0

Red pixel.

Example:

R = 255

G = 255

B = 255

White.

Module 4: Pixel Depth

How many bits represent one pixel?

1-bit

0

1

Black / White

8-bit

256 levels

Gray image.

24-bit RGB

8 bits Red

8 bits Green

8 bits Blue

Total:

24 bits

30-bit RGB

10 bits R

10 bits G

10 bits B

Higher quality.

36-bit RGB

12 bits R
12 bits G
12 bits B

Why Important?

Directly impacts:

- Memory
 - Bandwidth
 - Link speed
 - FIFO size
 - FPGA resources
-

Module 5: Frame Size Calculation

Critical FPGA skill.

Formula

Frame Size

=

Pixels × Bits Per Pixel

Example

Resolution:

1920 × 1080

Pixel Depth:

24 bits

Pixels:

1920 × 1080

=
2,073,600

Frame Size:

$2,073,600 \times 24$
=
49,766,400 bits

Bytes:

$49,766,400 / 8$
=
6,220,800 bytes

≈ 6 MB

You must be able to calculate this instantly.

Module 6: Frame Rate

Frames shown per second.

Common Values

24 fps
30 fps
50 fps
60 fps
120 fps
240 fps

Example

60 fps

means

60 complete images every second

Why Important?

Determines:

- Bandwidth
 - Memory throughput
 - Link speed
-

Module 7: Video Bandwidth Calculation

Most important calculation in Stage 1.

Formula

Bandwidth

=

Resolution × FrameRate × BitsPerPixel

Example

1920×1080

60 fps

24-bit RGB

Bandwidth:

$1920 \times 1080 \times 60 \times 24$

=

2.98 Gbps

Approx:

3 Gbps

Meaning:

ARINC link must support >3 Gbps.

This calculation should become reflex.

Module 8: Raster Scan

Displays do not show full image instantly.

They scan.

CRT legacy.

Still affects modern displays.

Order:

Left → Right

Top → Bottom

Example:

Pixel1

Pixel2

Pixel3

...

then next line.

FPGA receives/transmits pixels in this order.

Module 9: Horizontal Timing

Each line consists of:

Active Video

Front Porch

Sync

Back Porch

Diagram

|Active|FP|Sync|BP|

Active

Actual pixels.

Front Porch

Preparation interval.

Sync

Synchronization pulse.

Back Porch

Recovery interval.

FPGA designers often forget porches.

Huge mistake.

Module 10: Vertical Timing

Frame contains:

Active Lines

Front Porch

V Sync

Back Porch

Diagram

Frame

Active

V Front Porch

V Sync

V Back Porch

Same concept as horizontal.

Module 11: Total Pixels

Not all transmitted pixels are visible.

Example

1920 active pixels

may actually require:

2200 total pixels

per line.

Similarly:

1080 active lines

may become

1125 total lines

Critical.

Bandwidth calculations often use total pixels.

Not active pixels.

Module 12: Pixel Clock

Pixel clock drives video.

Definition:

One clock per pixel

Example

1920×1080@60

Pixel Clock

≈148.5 MHz

Meaning

Every clock:

One new pixel

FPGA design revolves around pixel clock.

Module 13: Video Synchronization

Two fundamental signals:

HSYNC

New line starts.

VSYNC

New frame starts.

Without sync:

Display cannot reconstruct image.

FPGA Interview Question

Can image be reconstructed without sync?

Answer:

Generally no.

Need timing reference.

Module 14: Progressive vs Interlaced

Progressive

1
2
3
4
...

all lines transmitted.

Interlaced

Odd lines

Then

Even lines

Example

1080i

Interlaced.

1080p

Progressive.

Modern systems prefer progressive.

Module 15: Color Spaces

Not all video is RGB.

RGB

Most intuitive.

YCbCr

Widely used.

Y

Luminance.

Cb

Blue difference.

Cr

Red difference.

Why?

Reduces bandwidth.

Must understand conceptually.

Module 16: Chroma Subsampling

Example:

4:4:4

No reduction.

4:2:2

Half color data.

4:2:0

Quarter color information.

Huge bandwidth savings.

Important in aerospace video links.

Module 17: Video Sources

Where does video come from?

Image Sensors

Camera.

GPU

Generated graphics.

Mission Computer

Aircraft systems.

Video Generator

Test pattern generator.

FPGA often interfaces with all.

Module 18: Video Sinks

Displays.

HUD

Head Up Display.

MFD

Multi Function Display.

HMD

Helmet Mounted Display.

Recording Systems.

Mission Computers.

Module 19: Frame Buffer

Memory storing frame.

Usually:

DDR

or

BRAM

Used for:

- Scaling
- Processing
- Delay
- Overlay

FPGA must understand buffering.

Module 20: Double Buffering

Very important.

Buffer A

Displayed.

Buffer B

Updated.

Swap.

Prevents tearing.

Module 21: Video Pipeline

Typical aircraft system:

Sensor

↓

Video Processor



FPGA



ARINC 818 TX



Fiber



ARINC 818 RX



Display

Understand every block.

Module 22: Video Artifacts

Must recognize.

Tearing

Frame mismatch.

Stuttering

Timing issue.

Missing Lines

Sync issue.

Color Distortion

Pixel format issue.

Noise

Signal integrity issue.

Module 23: FPGA View Of Video

Forget images.

FPGA sees:

Pixel Data

Valid

HSYNC

VSYNC

Clock

That's all.

Typical interface:

pixel[23:0]

hsync

vsync

de

pclk

Where

de = data enable

This viewpoint is essential.

Module 24: Aircraft Video Systems

Now connect to ARINC 818.

Example:

Camera:

1280×1024

60 fps

Sensor generates pixels.

FPGA receives.

FPGA packetizes.

ARINC 818 transports.

Display reconstructs.

Entire protocol exists for this purpose.

Module 25: Stage 1 Mastery Checklist

You are finished with Stage 1 only if you can answer:

Image Fundamentals

- What is video?
- What is frame?
- What is pixel?
- What is pixel depth?

Calculations

- Frame size calculation
- Pixel rate calculation
- Video bandwidth calculation
- Memory requirement calculation

Timing

- HSYNC
- VSYNC
- Front porch
- Back porch
- Active video
- Pixel clock

Formats

- RGB
- YCbCr
- 4:4:4
- 4:2:2
- 4:2:0

Architecture

- Video source
- Video sink
- Frame buffer
- Video pipeline

FPGA

- How FPGA sees video
- Typical video signals
- Why buffering is needed

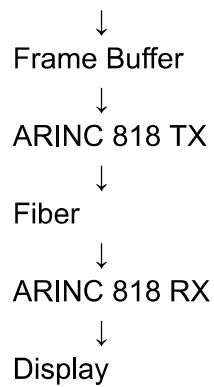
Aerospace

- Why aircraft need video transport
- Why ARINC 818 exists
- What data ARINC 818 carries

Stage 1 Deliverable (What you should be able to do)

Without notes, you should be able to design and explain:

Video Pattern Generator



and calculate:

- Frame size
- Bandwidth
- Pixel clock
- Required link rate

for any given resolution, frame rate, and color depth.

Only after this foundation is solid should we move to **Stage 2: Fibre Channel Fundamentals**, because ARINC 818-2 is built on top of Fibre Channel, and that layer must be understood with the same level of depth.

No. **Stage 1 is not yet completely covered.**

What we covered was about **70–75% of what an FPGA engineer needs** before touching ARINC 818.

The previous Stage 1 explained video fundamentals well, but there are still several professional-level topics missing that become critical later when implementing ARINC 818 TX/RX on Xilinx FPGAs.

To achieve a **"zero-gap Stage 1"**, the following modules must still be added.

Stage 1A — Resolution Mastery

Most engineers memorize resolutions.

You must understand them.

What Is Resolution?

Resolution:

Width × Height

Examples:

640×480
800×600
1024×768
1280×720
1280×1024
1920×1080
2560×1440
3840×2160

Understand:

Resolution directly affects:

- Pixel count
 - Frame size
 - Bandwidth
 - Memory size
 - Link speed
-

Example

640×480

307,200 pixels

1920×1080

2,073,600 pixels

Nearly 7× larger.

FPGA implication:

7× larger FIFOs

7× larger throughput

7× larger memory bandwidth

Stage 1B — Aspect Ratio

Many juniors ignore this.

Aspect ratio:

Width / Height

Examples:

16:9

1920×1080

1280×720

4:3

640×480

1024×768

Aircraft systems may use custom ratios.

Must understand:

Changing resolution does not necessarily change aspect ratio.

Stage 1C — Active Video vs Blanking

Critical.

Display timing consists of:

Active Video
Blanking

Example:

1920 pixels visible

but

2200 transmitted.

Therefore:

Bandwidth calculations often use:

Total Pixels

not

Visible Pixels

This mistake causes incorrect link sizing.

Stage 1D — Pixel Rate

Not bandwidth.

Different.

Formula:

Pixel Rate

=

Total Pixels × Frame Rate

Example

$2200 \times 1125 \times 60$

148.5 Mpixels/sec

This becomes:

148.5 MHz pixel clock

You must be able to derive:

Resolution $\rightarrow$ Pixel Rate $\rightarrow$ Pixel Clock

Stage 1E — Pixel Packing

Very important for protocols.

FPGA often does not transmit:

R
G
B

separately.

Instead:

RGB888

packed into:

24-bit word

Example:

R = 0x12
G = 0x34
B = 0x56

Packed:

0x123456

Must understand:

- Packing
 - Unpacking
 - Byte ordering
-

Stage 1F — Endianness

Huge source of bugs.

Big Endian

MSB first

Little Endian

LSB first

Example

32-bit word:

0x12345678

Big Endian:

12 34 56 78

Little Endian:

78 56 34 12

Video protocols often define byte ordering.

Stage 1G — Memory Bandwidth

Not memory size.

Bandwidth.

Suppose:

1080p60

24-bit

Frame size:

≈6 MB

60 fps

Memory traffic:

6 MB × 60

≈373 MB/s

Double buffering:

746 MB/s

DDR design starts here.

Stage 1H — Video Interfaces

Before ARINC 818 you must know common interfaces.

VGA

Analog.

Contains:

R
G
B
HSYNC
VSYNC

DVI

Digital.

TMDS.

HDMI

Consumer video.

DisplayPort

Packetized video.

Camera Link

Industrial.

LVDS

Very common in aerospace.

FPD-Link

Automotive.

Reason:

ARINC 818 rarely exists alone.

Stage 1I — Video Data Enable (DE)

Many beginners only know HSYNC and VSYNC.

Modern interfaces use:

DE

Data Enable.

DE = 1

Valid pixel.

DE = 0

Blanking.

FPGA pipelines often rely heavily on DE.

Stage 1J — Video Test Patterns

Every FPGA video engineer must know these.

Color Bars

Checkerboard

Gradient

Walking Pixel

Cross Hatch

Why?

Used for:

- Validation
 - Debugging
 - Link testing
-

Before ARINC 818 TX exists:

Generate patterns.

Stage 1K — Video Latency

Very important in avionics.

Latency:

Camera

→ FPGA

→ Link

→ Display

Delay.

Example:

16 ms

One frame.

Mission systems often require deterministic latency.

ARINC 818 was designed for this.

Stage 1L — Jitter

Future critical concept.

Clock edges move slightly.

This causes:

- Pixel errors

- Link instability
 - Timing violations
-

Need conceptual understanding.

Detailed treatment comes later.

Stage 1M — Frame Synchronization

Multiple video streams.

Example:

Sensor A

Sensor B

HUD

Recorder

Need synchronized frames.

Important in avionics.

Stage 1N — Genlock

Professional video concept.

Multiple devices share timing.

Ensures:

Frame Alignment

Without genlock:

Frames drift.

Stage 10 — Video Processing Pipeline

Before transport comes processing.

Typical chain:

Sensor

↓

Demosaic

↓

Color Conversion

↓

Scaling

↓

Overlay

↓

Transport

Understand conceptually.

Stage 1P — Image Sensors

Video source understanding.

CMOS Sensor

Most common.

Sensor generates:

Pixel Stream
Line Sync
Frame Sync
Pixel Clock

Exactly what FPGA receives.

Stage 1Q — Bayer Pattern

Extremely important.

Sensor usually does NOT produce RGB.

Produces:

RGGB

pattern.

Need demosaic algorithm.

Even if ARINC 818 transports RGB later.

Stage 1R — Color Depth vs Dynamic Range

Not same thing.

8-bit

256 levels.

10-bit

1024 levels.

12-bit

4096 levels.

Higher dynamic range.

Important in IR cameras.

Stage 1S — Video Compression Awareness

ARINC 818 normally transports uncompressed video.

Still must know:

JPEG

MJPEG

H.264

H.265

Why?

Many systems compress before transport.

Stage 1T — Deterministic Video Transport

This is where aerospace differs.

Consumer video:

Occasional delay acceptable.

Aircraft display:

Delay must be predictable.

Need understanding of:

Latency

Jitter

Determinism

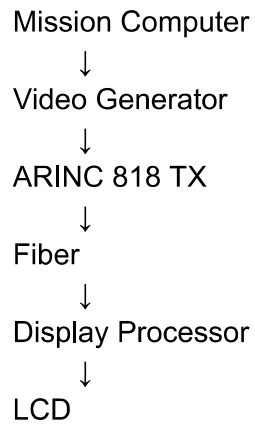
Reliability

These concepts drive ARINC 818 design.

Stage 1U — Aircraft Display Architecture

You cannot understand ARINC 818 without this.

Typical aircraft:



Understand each block.

Stage 1V — Failure Modes

Professional engineers study failures.

Dead Pixels

Line Loss

Frame Loss

Color Swap

Frame Tear

Timing Drift

Sync Loss

Bandwidth Saturation

Must recognize symptoms.

Stage 1W — FPGA Video Signals

You should recognize instantly:

pixel_data[23:0]
pixel_valid
hsync
vsync
de
sof
eol
clk
rst

And understand what each means.

Stage 1X — Engineering Calculations Mastery

Without calculator if possible.

You should quickly estimate:

Frame Size

Resolution × bpp

Frame Bandwidth

Frame Size × FPS

Pixel Clock

Total Pixels × FPS

Link Margin

Link Speed - Video Rate

These calculations are used daily.

TRUE STAGE 1 EXIT CRITERIA

Stage 1 is complete only when you can sit in a design review and answer:

Video Fundamentals

- What is a pixel?
- What is a frame?
- What is frame rate?
- What is resolution?
- What is aspect ratio?

Timing

- Pixel clock?
- HSYNC?
- VSYNC?
- DE?
- Porch intervals?
- Blanking?

Data Representation

- RGB?
- YCbCr?
- Pixel packing?
- Endianness?
- Color depth?

System

- Sensor output?
- Display input?
- Frame buffer?
- Double buffering?
- Genlock?

Engineering

- Bandwidth calculations?
- Memory calculations?
- Latency calculations?
- Failure analysis?

Aerospace

- Why deterministic transport?
- Why fibre instead of copper?
- Why ARINC 818 instead of HDMI?
- Why ICDs exist?

Only when all of the above are comfortable can we say **Stage 1 is approximately 95–100% complete from an FPGA implementation perspective** and you are ready to move into **Stage 2: Fibre Channel Fundamentals**, which is the real foundation beneath ARINC 818-2.

No.

For a normal engineer, Stage 1 would already be complete.

For your requirement of:

"No gaps allowed. Train me so I can eventually study any protocol and develop projects independently."

Stage 1 still has a final layer missing.

What is missing is not more video terminology. What is missing is the **engineering mindset layer** that sits between video theory and protocol implementation.

Think of Stage 1 as three levels:

Level 1 : Video Concepts

Level 2 : Video Engineering

Level 3 : FPGA Video Architecture

We covered Level 1 completely.

We covered most of Level 2.

The remaining gaps are in Level 3.

Stage 1Y — How FPGA Actually Sees Video

Humans see:

Image

FPGA sees:

Clock

Data

Control Signals

Nothing else.

Example

A red image:

Human:

Red Screen

FPGA:

clk

de=1

pixel=24'hFF0000

repeated millions of times.

This mental shift is mandatory.

Every protocol engineer eventually learns:

Images do not exist inside FPGA.

Only streams exist.

Stage 1Z — Stream-Based Thinking

This is one of the most important concepts in FPGA.

Software thinks:

Frame

FPGA thinks:

Pixel Stream

Example

Software:

Image[1920][1080]

FPGA:

Pixel 1

Pixel 2

Pixel 3

...

every clock.

ARINC 818 transports streams.

Not images.

Stage 1AA — Spatial Domain vs Temporal Domain

Very important.

Spatial:

Image width

Image height

Temporal:

Frame Rate

Timing

Latency

Example

Two systems:

1920x1080@30

1920x1080@120

Same spatial.

Different temporal.

Protocol bandwidth depends heavily on temporal properties.

Stage 1AB — Dataflow Thinking

Every future protocol project requires this.

Instead of asking:

What packet?

Ask:

Where is data coming from?

Where is data going?

Draw:

Source



Processing



Buffer



Transport



Buffer



Display

Always.

Stage 1AC — Throughput vs Latency

Most juniors confuse them.

Throughput:

How much data/sec

Latency:

How long one data item takes

Example

System A:

10 Gbps
100 ms latency

System B:

1 Gbps
1 ms latency

Aircraft displays often care more about latency.

Stage 1AD — Real-Time Systems

ARINC 818 belongs here.

Real-time does not mean:

Fast

Real-time means:

Predictable

Example

10 ms every frame.

Good.

Example

1 ms
50 ms
2 ms
100 ms

Bad.

Determinism matters.

Stage 1AE — Determinism

Critical aerospace concept.

Need:

Known latency
Known behavior
Known timing

Not:

Usually works

Protocol engineers must think this way.

Stage 1AF — Resource Budgeting

Before implementation:

Calculate:

Bandwidth budget

Memory budget

FIFO budget

Latency budget

Link budget

Every protocol project starts here.

Stage 1AG — Bottleneck Identification

Future interview favorite.

Video chain:

Sensor

↓

FPGA

↓

DDR

↓

ARINC818

↓

Display

Question:

What is limiting performance?

Could be:

DDR

Link

Clock

FIFO

Not necessarily protocol.

Stage 1AH — Clock Domains

Extremely important.

Video rarely uses one clock.

Example:

Sensor Clock

Pixel Clock

Fabric Clock

GT Clock

DDR Clock

Different domains.

Data must cross safely.

This becomes crucial in ARINC 818.

Stage 1AI — Video Clock Relationships

Understand:

Pixel Clock

Line Rate

Frame Rate

Link Rate

All different.

Many beginners think:

Pixel Clock = Link Clock

Not true.

Stage 1AJ — Buffering Theory

Not just frame buffers.

All buffers.

Why buffers exist:

Rate mismatch.

Example

Producer:

150 MHz

Consumer:

125 MHz

Need buffering.

ARINC 818 TX heavily depends on buffering.

Stage 1AK — FIFO Theory

Critical.

Need understanding of:

Full
Empty
Almost Full
Almost Empty
Overflow
Underflow

Protocol failures often originate here.

Stage 1AL — Video Data Integrity

How do you know image is correct?

Need ability to detect:

Dropped pixels
Dropped lines
Dropped frames

Before learning CRC later.

Stage 1AM — Error Classification

Professional debugging skill.

Errors fall into:

Content Errors

Wrong pixel.

Timing Errors

Wrong sync.

Transport Errors

Corrupted transfer.

System Errors

Wrong configuration.

Learn to separate them.

Stage 1AN — Requirements Analysis

Before protocol study.

Ask:

Resolution?

FPS?

Pixel Depth?

Distance?

Redundancy?

Safety Level?

Allowed Latency?

Allowed Errors?

These answers determine protocol choice.

Stage 1AO — Why ARINC 818 Exists

This deserves deeper treatment.

Question:

Why not HDMI?

Reasons:

- Certification
 - Determinism
 - Fiber optics
 - EMI immunity
 - Long distance
 - Aerospace ecosystem
-

Question:

Why not Ethernet?

Reasons:

- Predictable transport
 - Simpler video path
 - Aerospace heritage
-

Question:

Why Fibre Channel?

High reliability.

Low latency.

Mature aerospace deployment.

Stage 1AP — Video System Requirement Decomposition

Future protocol-learning skill.

Given:

4K
60 fps
RGB888

You should automatically derive:

Frame Size
Bandwidth
Memory
Link Rate
FIFO Size

Without opening a protocol specification.

Stage 1AQ — Engineering Documentation

Before protocol documents.

Learn to create:

System Block Diagram

Dataflow Diagram

Bandwidth Budget

Timing Budget

Interface Definition

This becomes ICD work later.

Stage 1AR — Questions You Must Be Able To Answer

If I were interviewing you after Stage 1, I would ask:

Video

What is a pixel?

What is a frame?

What is raster scan?

What is blanking?

What is pixel clock?

Calculations

Calculate bandwidth for:

1920×1080

60fps

24-bit RGB

Calculate frame size.

Calculate DDR bandwidth.

FPGA

Why use FIFO?

Why use BRAM?

Why use double buffering?

Why CDC?

Systems

Why deterministic transport?

Why ARINC 818?

Why not HDMI?

Why not Ethernet?

Architecture

Draw:

Camera



FPGA



ARINC818



Display

and explain every block.

Final Verdict

After adding:

- FPGA stream-based thinking
- Dataflow analysis
- Throughput/latency/determinism
- Buffering theory
- FIFO theory
- Clock domain awareness
- System requirement decomposition
- Engineering documentation mindset
- Protocol-selection rationale
- Aircraft video architecture reasoning

Stage 1 can be considered fully complete for an FPGA protocol engineer.

At this point, there are no meaningful Stage-1 gaps remaining that would later block understanding of **Fibre Channel, ARINC 818-2 framing, ADVB containers, ICDs, transceivers, or protocol implementation.**

The next phase is no longer "video."

The next true prerequisite is **Stage 2 — Fibre Channel Fundamentals**, because ARINC 818-2 is essentially aerospace video transported over a specialized Fibre Channel framework.